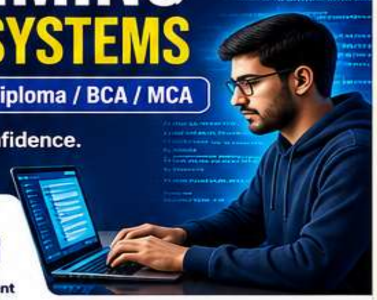
C PROGRAMMING
FOR EMBEDDED SYSTEMS

COMPLETE COURSE FOR B.Tech / M.Tech / Diploma / BCA / MCA

Build Strong Fundamentals. Code with Confidence.
Design the Future.



From Basics to Advanced | Embedded Focused | Hands-on Learning | Real-world Applications | Interview Preparation | Placement Oriented

BATCH TIMING

3Days/Week
Sat-Sun : 9AM-11AM
Workingday : 9:30PM-11PM
(Flexible)

COURSE FEE

₹ 3000
(Full Course)

TARGET STUDENTS

- B.Tech / M.Tech / Diploma Students
- BCA / MCA Students
- Beginners in Programming
- Students Preparing for Placements
- Aspiring Embedded System Engineers

WHY JOIN THIS COURSE?

- Complete Syllabus from Basics to Advanced Level
- Industry-Oriented Teaching
- Practical & Hands-on Approach
- Real Embedded Examples

CONTACT

+91 9628416763

COURSE HIGHLIGHTS

Strong C Fundamentals | Pointers & Memory Mastery | Embedded C Programming | MCU & Peripheral Understanding | Real Time Applications | Interview & Career Support

WHAT YOU WILL LEARN?

- Core C Programming Concepts
- Memory Management & Pointers
- Data Structures using C
- File Handling
- Bit Manipulation & Embedded Concepts
- Microcontroller & Peripheral Programming
- RTOS Concepts (Basics)
- Coding Standards & Best Practices
- Debugging Techniques
- Mini Projects & Real Time Applications

MODE OF LEARNING

Live Online Classes
+ Doubt Solving Sessions

WHO WILL BENEFIT?

Students who want to build a strong career in Embedded Systems, Firmware Development, Software Development and Core Programming.

PREREQUISITES

Basic Logic Knowledge
(C Language – No Prior Coding Experience Required)

COMPLETE DAY-WISE SYLLABUS (13 MODULES)

MODULE 1 – INTRODUCTION TO C PROGRAMMING

Day 1	Introduction to Programming, C & Its Applications
Day 2	Structure of C Program, Compilation Process
Day 3	Data Types, Variables, Constants
Day 4	Input / Output Functions
Day 5	Operators in C
Day 6	Simple Programs
Day 7	Practice & Doubt Session

MODULE 2 – CONTROL STATEMENTS

Day 8	if Statement
Day 9	if-else & else-if Ladder
Day 10	switch-case Statement
Day 11	while Loop
Day 12	do-while Loop
Day 13	for Loop
Day 14	Nested Loops & Patterns

MODULE 3 – FUNCTIONS

Day 15	Functions – Basics
Day 16	Function Declaration & Definition
Day 17	Call by Value
Day 18	Call by Reference
Day 19	Recursion
Day 20	Practice Programs
Day 21	Practice & Doubt Session

MODULE 4 – ARRAYS & STRINGS

Day 22	1D Arrays
Day 23	2D Arrays
Day 24	Array Applications
Day 25	Strings in C
Day 26	String Functions
Day 27	Character Arrays vs Strings
Day 28	Practice & Doubt Session

MODULE 5 – POINTERS

Day 29	Introduction to Pointers
Day 30	Pointer Arithmetic
Day 31	Pointers and Arrays
Day 32	Pointers to Functions
Day 33	Double Pointers
Day 34	Void Pointers
Day 35	Practice & Doubt Session

MODULE 6 – STRUCTURES, UNION & ENUM

Day 36	Structures
Day 37	Nested Structures
Day 38	Array of Structures
Day 39	Pointer to Structure
Day 40	Union
Day 41	enum & typedef
Day 42	Practice & Doubt Session

MODULE 7 – DYNAMIC MEMORY ALLOCATION

Day 43	Introduction to Dynamic Memory
Day 44	malloc()
Day 45	calloc()
Day 46	realloc()
Day 47	free() and Memory Management
Day 48	Memory Leaks & Best Practices
Day 49	Practice & Doubt Session

MODULE 8 – FILE HANDLING

Day 50	Files in C
Day 51	Opening & Closing Files
Day 52	Reading a File
Day 53	Writing to a File
Day 54	fprintf() and fscanf()
Day 55	Binary Files
Day 56	Practice & Doubt Session

MODULE 9 – BITWISE PROGRAMMING

Day 57	Binary Number System
Day 58	Bitwise Operators
Day 59	Bit Masking Techniques
Day 60	Setting, Clearing & Toggling Bits
Day 61	Register Level Programming Basics
Day 62	Practice Programs
Day 63	Practice & Doubt Session

MODULE 10 – EMBEDDED C PROGRAMMING

Day 64	Introduction to Embedded Systems
Day 65	Embedded C vs General C
Day 66	Volatile Keyword
Day 67	Memory Mapped I/O
Day 68	Writing Embedded Style Code
Day 69	Coding Standards (MISRA C Basics)
Day 70	Practice & Doubt Session

MODULE 11 – MICROCONTROLLER BASICS

Day 71	Introduction to Microcontrollers
Day 72	ARM Cortex-M Overview
Day 73	Memory Map of MCU
Day 74	GPIO Programming
Day 75	UART Communicating
Day 76	SPI, I2C, CAN Basics
Day 77	Practice & Doubt Session

MODULE 12 – REAL-TIME & RTOS CONCEPTS

Day 78	Interrupts in Embedded Systems
Day 79	Timers & Watchdog
Day 80	RTOS Concepts (Basics)
Day 81	Task, Scheduling, Priority
Day 82	Debugging Techniques
Day 83	GDB Basics
Day 84	Practice & Doubt Session

MODULE 13 – DEBUGGING & INTERVIEW PREPARATION

Day 85	Common C Mistakes
Day 86	Segmentation Fault
Day 87	Buffer Overflow Basics
Day 88	Debugging Techniques
Day 89	C Interview Questions
Day 90	Embedded Interview Questions
Day 91	Practice & Doubt Session

MINI PROJECTS INCLUDED

- Calculator Application
- Student Management System
- File Handling Project
- Register Level Programming Simulation
- Bit Masking Applications
- Embedded Peripheral Communication
- Mini Embedded Projects

COURSE OUTCOMES

- Write C programs confidently
- Deep understanding of Pointers & Memory
- Build strong programming logic
- Understand Embedded Systems Basics
- Learn Register-Level Programming
- Understand Peripheral Communication
- Prepare for Technical Interviews
- Ready for Internships & Placements

Code Today,
Build Tomorrow,
Lead the Future!

SPECIAL FEATURES

Live Sessions | Industry Oriented Content | Practical Coding Approach | Real Time Examples

Learn From Working Professional

GET IN TOUCH

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C Programming for Embedded Systems

Complete Course Syllabus

Course Details

Course Name

C Programming for Embedded Systems

Course Timing

 3days / week
Sat-Sun: - 9:00 AM – 11 AM

Working day: - 9:30PM-11PM

(Flexible)

Course Fee

₹ 3000

Pay only ₹ 500 to start, ₹ 2500 after completing module 1

Don’t like course, may exit during module 1 and get ₹ 500 refund.

Mode

Online / Offline Live Classes

Batch strength

20 student / batch

Target Students

- B.Tech/M.Tech Students
- Diploma Students
- ECE / EEE / CSE / ISE Students
- BCA/MCA
- Beginners in Programming
- Embedded Systems Aspirants

- Placement Preparation Students

Course Objective

This course is designed to build:

- Strong programming fundamentals
- Logical problem-solving skills
- Deep understanding of C language
- Embedded systems programming basics
- Interview preparation skills
- Real-time programming concepts

Students will learn from:

- ✔ Basics to Advanced C
- ✔ Embedded C Concepts
- ✔ Real-Time Programming Basics
- ✔ Industry-Oriented Programming Practices

MODULE 1 — INTRODUCTION TO C PROGRAMMING

Topics Covered

- Introduction to Programming
- Software vs Hardware
- Programming Languages
- Why C Language?
- Structure of C Program
- Compilation Process
- Header Files
- Keywords & Identifiers
- Variables & Constants
- Data Types
- Input / Output Functions
- Format Specifiers

Practical Programs

- Hello World
- Calculator Program
- Temperature Converter
- User Input Programs

MODULE 2 — OPERATORS & CONTROL STATEMENTS

Topics Covered

- Arithmetic Operators
- Relational Operators
- Logical Operators
- Assignment Operators
- Increment & Decrement
- Operator Precedence
- if Statement
- if-else
- Nested if
- else-if Ladder
- switch-case
- while Loop
- do-while Loop
- for Loop
- Nested Loops

Practical Programs

- Even/Odd
- Largest Number
- Grade System
- Multiplication Table
- Prime Number
- Fibonacci Series
- Pattern Programs

MODULE 3 — FUNCTIONS

Topics Covered

- Function Basics
- Function Declaration
- Function Definition
- Function Calling
- Return Type
- Arguments & Parameters
- Call by Value
- Call by Reference
- Recursive Functions
- Storage Classes

- auto
- static
- extern
- register

Practical Programs

- Calculator using Functions
- Recursive Programs
- Menu Driven Applications

MODULE 4 — ARRAYS & STRINGS

Topics Covered

Arrays

- 1D Arrays
- 2D Arrays
- Matrix Operations
- Array Initialization

Strings

- String Basics
- String Functions
- String Manipulation
- Command Line Arguments

Practical Programs

- Matrix Addition
- Matrix Multiplication
- String Reverse
- Palindrome String
- Word Count

MODULE 5 — POINTERS

Topics Covered

- Introduction to Pointers
- Address Operators
- Pointer Declaration
- Pointer Arithmetic
- Pointers & Arrays

- Pointers to Functions
- Double Pointers
- Void Pointers
- Null Pointers

Important Concepts

- Stack Memory
- Heap Memory
- Memory Layout
- Memory Access

Practical Programs

- Pointer Arithmetic
- Array Traversal using Pointers
- Function Pointer Examples

MODULE 6 — STRUCTURES, UNION & ENUM

Topics Covered

- Structures
- Nested Structures
- Array of Structures
- Pointer to Structure
- typedef
- Union
- Enum

Practical Programs

- Student Database
- Employee Record System
- State Machine Examples

MODULE 7 — DYNAMIC MEMORY ALLOCATION

Topics Covered

- malloc()
- calloc()
- realloc()
- free()

Important Concepts

- Heap Allocation
- Dynamic Memory
- Memory Leaks
- Dangling Pointers
- Best Practices

Practical Programs

- Dynamic Array Creation
- Dynamic Data Storage

MODULE 8 — FILE HANDLING

Topics Covered

- File Basics
- fopen()
- fclose()
- fprintf()
- fscanf()
- fread()
- fwrite()

File Modes

- Read
- Write
- Append
- Binary File Operations

Practical Programs

- Student Database using Files
- Log File Generator

MODULE 9 — BITWISE PROGRAMMING

Topics Covered

- Binary Number System
- Hexadecimal Basics
- Bitwise AND
- Bitwise OR
- XOR

- NOT Operator
- Shift Operators
- Bit Masking
- Setting/Clearing/Toggling Bits

Embedded Relevance

- Register Manipulation
- Hardware Control
- Peripheral Configuration

Practical Programs

- Bit Manipulation Programs
- Register Simulation

MODULE 10 — EMBEDDED C PROGRAMMING

Topics Covered

- Introduction to Embedded Systems
- Embedded Software Architecture
- Bare Metal Programming
- Register Level Programming
- Memory Mapped IO
- Volatile Keyword
- Embedded Coding Standards
- MISRA C Basics

Practical Programs

- GPIO Register Simulation
- Embedded Driver Logic

MODULE 11 — MICROCONTROLLER BASICS

Topics Covered

- Introduction to Microcontrollers
- ARM Cortex-M Basics
- Flash & RAM
- CPU Registers
- Stack & Memory
- Interrupt Basics

Peripheral Basics

- GPIO
- UART
- SPI
- I2C
- CAN Communication Basics

Practical Programs

- Peripheral Register Access
- UART Communication Logic

MODULE 12 — REAL-TIME & RTOS CONCEPTS

Topics Covered

- What is RTOS?
- Tasks & Scheduling
- Priority Concepts
- Context Switching
- Interrupt Latency
- Watchdog Timer
- Real-Time Systems
- Debugging Basics

Practical Concepts

- RTOS Demonstration
- Timing Analysis
- Task Scheduling Examples

MODULE 13 — DEBUGGING & INTERVIEW PREPARATION

Topics Covered

- Debugging Techniques
- Breakpoints
- Watch Window
- printf Debugging
- Common Programming Errors
- Segmentation Fault
- Infinite Loops

- Buffer Overflow Basics

Interview Preparation

- Common C Interview Questions
- Embedded Interview Questions
- Logic Building Questions
- Placement Preparation

MINI PROJECTS INCLUDED

Students will work on:

- ✔ Calculator Application
- ✔ Student Management System
- ✔ File Handling Project
- ✔ Register Level Programming Simulation
- ✔ Bit Masking Applications
- ✔ Embedded Peripheral Logic
- ✔ Mini Embedded Projects

KEY LEARNING OUTCOMES

After completing this course students will be able to:

- ✔ Write C programs confidently
- ✔ Understand pointers & memory deeply
- ✔ Build strong programming logic
- ✔ Understand embedded system basics
- ✔ Learn register-level programming
- ✔ Understand peripheral communication basics
- ✔ Prepare for technical interviews
- ✔ Start Embedded Systems development

WHY JOIN THIS COURSE?

- ✔ Beginner Friendly Teaching
- ✔ Industry-Oriented Examples
- ✔ Embedded Systems Focus
- ✔ Practical Coding Sessions
- ✔ Real-Time Examples
- ✔ Learn from Working Professional

CONTACT DETAILS

- Contact Number: +91 9628416763
- Location / Mode: Online Live Session